Fintech and Entrepreneurship: An Assessment Model to Evaluate Policy Instruments for Fintech Adoption by Small and Medium Enterprises

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***Abstract*—Small and medium enterprises (SMEs) are the engines that drive economic development. However, they face inferior conditions and challenges when it comes to their access to broad financing channels. Hence, financial technologies (Fintech) solutions offer promising potential for improving SMEs' access to finance. While researchers and practitioners have been promoting Fintech as a potential financial safeguard for SMEs' needs, evidence shows an inadequate adoption rate of SMEs to such solutions and a lack of a comprehensive view of the Fintech adoption. Therefore, this research aims to provide a comprehensive examination of Fintech policy instruments and analyze their effectiveness in increasing the adoption of Fintech by SMEs through evaluating the essential policy targets impacting the adoption of Fintech and assessing their weights and priorities. This research is built upon a comprehensive literature review and experts' insights. The hierarchical decision modeling methodology is used to identify the relative importance of those factors proposing a policy evaluation tool to assess the effectiveness of policy instruments on increasing Fintech adoption. This research presents the identification of 17 distinct policy targets that fall within four main perspectives along with their relative weights. To test the practicality and value of the research, a case study of the policy instrument effectiveness, the Saudi Arabian regulatory sandbox, was conducted to illustrate how the model could be used to identify the policy instrument's actual performance in terms of influencing SMEs adoption of Fintech, identify the instrument's strengths and weaknesses, and offer recommendations on how to improve to increase the policy instrument's effectiveness in increasing the adoption of Fintech by SMEs.**

***Index Terms*—Decision making in technology management, entrepreneurship, Fintech, small and medium enterprises (SMEs), technology assessment, technology entrepreneurship.**

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I. Introduction

SMALL and medium enterprises (SMEs) are the backbone of economies around the world [1]. It has been estimated that SMEs represent 95% of a firm's population and account for 60% to 70% of employment [2]. However, SMEs face challenges of many sorts. The literature highlights the financial challenges that are associated with SMEs include high transaction costs, information asymmetry, and confronting higher risks in their operations [3]. Consequently, these constraints cause SMEs' growth to be delayed. Furthermore, SMEs do not have access to broad financing channels where their financing circumstances are unfavorable, especially because of their size and fragile nature, where they face various drawbacks and challenges and have a high expectancy of failure [4]. Hence, Fintech solutions offer the possibility for improving SME access to finance, extending more accessible and available services, and reducing transaction costs [5]. These tools can offer a valuable opportunity for ventures that are too small in size, and involve a great deal of risk, or serve a social purpose [6], [7]. Fintech proposes promising solutions for such financial challenges. The evolution of Fintech solutions has resulted in a growth in the access and convenience of financial services, for both individual consumers and SMEs [8]. However, with all the benefits projected by Fintech, the adoption rate of Fintech by SMEs has been considerably low compared to the rate of consumer adoption. According to the Fintech Adoption Index conducted by Ernst and Young Global in 2019 [9], only 25% of SMEs are adopting Fintech globally compared to 60% of global consumer adoption. More importantly, the literature has emphasized the significant role of government support in the acceleration of Fintech adoption by SMEs [10].

While researchers and policy makers have been suggesting Fintech as a potential financial safeguard for SMEs, evidence shows inadequate adoption of SMEs to such solutions. More importantly, when implementing new policies, it is important for governments to prove that their decisions are informed, their policies are supported by evidence, and they have laid reasonable

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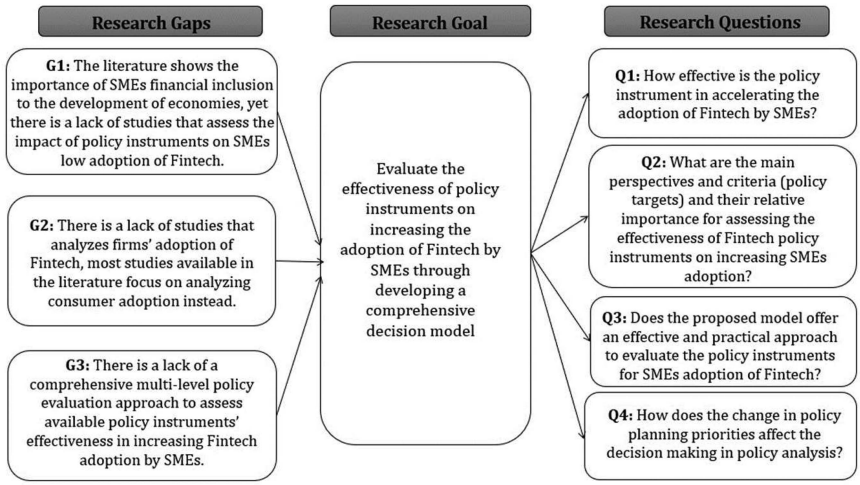


Fig. 1. Gap analysis.

expectations concerning their policy selections. Therefore, policy makers need policy evaluation tools to ensure the fulfillment of these objectives as well as eliminate the possibility of policy failure [11]. Evaluating performance and outcomes enables policymakers to grasp a deeper understanding of underlying complications a policy may encounter and, hence, aid them make knowledgeable decisions regarding whether it is worthwhile to keep the current policy going, or whether it would be more viable to create a new one instead. More importantly, currently, there are no models or tools designed to evaluate and assess the impact of the available policy instruments in increasing SMEs adoption of Fintech by considering multiple perspectives to measure the instruments' effectiveness. Because of Fintech's intercrossed domains and complexity of its integrated systems, multicriteria decision making (MCDM) analysis methods offer a suitable solution for such complex evaluations [12], [13], [14].

Based on the literature review on entrepreneurship and the policy instruments of Fintech adoption that assess facilitating the existing barriers preventing SMEs from adopting such valuable technologies and solutions, the following gaps were identified in the literature.

GAP 1: Lack of Available Assessment of Low Fintech Adoption: The literature emphasizes the importance of SMEs financial inclusion for the development of the economy and how Fintech can be an effective solution for such matters. In addition, while researchers and policy makers have been suggesting Fintech as a potential financial safeguard for SMEs, evidence shows inadequate adoption of SMEs to such solutions. More importantly, there is a lack of studies that assess the low adoption rate of Fintech by SMEs. Hence, there is a need for a study that evaluates available policy instruments directed toward Fintech adoption, by assessing their effectiveness in achieving that objective while taking into consideration the various influencing factors relative to the assessment and proposing a model that facilitates that assessment. The study of the assessment of policy instruments for the adoption of Fintech from different perspectives would greatly aid in understanding the broad state of such gap, and identify the critical factors that would influence and enable the adoption of Fintech by SMEs and entrepreneurs.

GAP 2: Limited Perspective: In addition, there is a lack of studies that focus on firms' adoption of Fintech, where most studies available in the literature focus on the consumer's adoption instead. Thus, there is a need to identify and define evaluation criteria to be used in developing an assessment model for policy instruments designed to increase Fintech adoption by firms, more specifically, SMEs.

GAP 3: Lack of a Comprehensive Hierarchical Model: Lack of Approaches: Moreover, there is a lack of a comprehensive multilevel approach that analyzes the adoption and its associated factors and barriers. Most literature focuses on the adoption from a limited perspective, ignoring the importance of a multilevel assessment that can investigate the adoption case and generate more novel and comprehensive conclusions.

The objective of this research is, therefore, to evaluate the effectiveness of available policy instruments on increasing the adoption of Fintech by SMEs. It will provide a comprehensive framework for policy makers to help simplify the understanding of the factors affecting the adoption of financial technologies by SMEs. It will present a hierarchical decision-making model (HDM) as a policy assessment tool for policy decision makers to analyze policy instruments and create the ideal Fintech policy portfolio, through the identification of the critical and challenging factors that need to be accurately addressed in a policy instrument to enable and increase the adoption of Fintech. The research provides the following output.

1) Identification of the perspectives and criteria for evaluating the effectiveness of the policy instruments on increasing the adoption of Fintech by SMEs.
2) Identification of the relative importance of each perspective and criterion to the assessment process.
3) A tool for policy makers to assess the effectiveness of their policy instruments on the adoption in order to overcome the weaknesses in the existing policy instruments.
4) Assessment of the effectiveness and feasibility of the developed model for evaluating the effectiveness of policy instruments on increasing Fintech adoption by SMEs.

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II. Literature Review

A. Small and Medium Enterprises

In this era of globalization, SMEs are considered essential to any country's advancement [15], [16], [17]. It is widely acknowledged that SMEs contribute significantly to an economy [18]. Regardless of their small-scale output and comparatively high production expenses, SMEs contribute substantially to job creation, poverty alleviation, and economic growth [1]. The roughly 400 million SMEs are the backbone of economies around the world, especially in developing and emerging countries [2]. SMEs appear to have an advantage over large firms due to their fast and flexible decision-making processes. In general, small firms' relative strengths are mainly behavioral, such as entrepreneurial dynamism, flexibility, efficiency, and quick decision making. On the other hand, the strengths of large firms include economies of scale, scope, marketing skills, and financial and technological resources. Large firms, backed up with more resources, react better to trade impediments than SMEs, which provides them a competitive advantage in global markets [19], [20].

B. Fintech

Fintech is defined by the Financial Stability Board as "technology-enabled innovation in financial services that could result in new business models, applications, processes or products with an associated material effect on the provision of financial services" [8]. This term encompasses a large scope of practices ranging from data security to financial service deliveries [12]. The term's origin can be traced to the early 1990s and referred to the "Financial Services Technology Consortium," a project started by Citigroup to enable technological cooperation attempts [21]. However, it is only since 2014 that the sector has drawn the attention of regulators, industry participants, and consumers, as well. Fintech includes digital innovations and technology-empowered business model innovations in the financial industry [22]. These innovations can disrupt the current industry structures and relax boundaries within the industry, enable strategic disintermediation, renovate the way firms make and carry products and services currently, offer entrepreneurs new access opportunities to the market, and increase equity access to financial services.

Fintech has been considered a distinctive taxonomy that primarily defines the financial technology sectors in a large range of operations for firms or organizations, which mostly highlight the enhancement of the quality of the service by using information technology (IT) capabilities [12]. Since the investment has been constantly growing, this has been fueling the growth of Fintech to progress on technological innovations in several areas, such as trust management [23], [24], cloud computing [25], and data analytic techniques [26], [27]. It has been reported that Fintech has become one of the main investments among most competitive financial companies [28]. As the prospects of using Fintech have increased dramatically, a great challenge to its adoption and planning has been triggered, mainly because of the intercrossed domains, the complexity of its integrated systems, and unique demands. Hence, there is an urgent demand for both academics and professionals to deliver correct and updated knowledge of Fintech [12].

C. Policy Instruments for Fintech Adoption

Given the importance of SMEs in emerging countries' private sectors, regulators and policy makers have been focusing on SMEs' financing as a priority zone for policy assistance and involvement. Alternative finance activities that are offered online have been gradually incorporated into the SME finance policy initiatives. Fintech solutions offer the possibility for improving SME access to finance, extending more accessible and available services, more efficient credit risk assessments, and reduced transaction costs [5]. These tools can offer a valuable opportunity for ventures that are too small in size, and involve a great deal of risk, or serve a social purpose urging regulators' involvement [6], [7]. In addition, it has been proven that financial aid is most effective once it is offered in combination with nonfinancial aid, such as mentoring, counseling, consulting, or general financial education [29], since SMEs occasionally encounter skills gaps in addition to the financing gap [30].

According to the literature, Fintech, BigTech, and digitalization are rapidly transforming the financial services industry [31]. These innovations hold the promise of increasing financial inclusion for the benefit of individuals, businesses, and the overall macroeconomy. Furthermore, each country needs to create an environment where regulators of such priorities can cooperate with each other. Also, governments need to create an enabling political and legal environment that prioritizes digital financial services to permit the flow of remittances into the country [32]. This may require constructing structures that enable more formal or organized coordination among various forms of regulators [31]. Hence, regulators have to realize the value of making digital financial services a priority and present a clear route toward regulating Fintech and digital financial services providers [32]. The evolution of digitalization has also created the need for modernizing the antitrust regulations and policies, which has already started happening in multiple countries [31]. Moreover, many jurisdictions have restored their regulatory and supervisory structure to enable digitalization to occur in the financial sector. Hence, there has been an urgent need for governments to adopt policies that increase Fintech support [32].

D. Fintech Regulatory Sandboxes

Fintech regulations aim to create a policy framework that would promote and support disruptive innovation to improve financial inclusion and economic growth as well as defend the security and reliability of the banking system and financial steadiness in general [33]. According to Brummer and Yadav, market integrity, rules simplicity, and financial innovations are the basis for achieving these objectives. Yet, there are compromises and possible interplay that may hinder one another. Generally, regulators aim to provide clear guidelines, preserve market integrity, and promote Fintech innovations, as they design the Fintech regulations, yet it is more likely that they would attain only two out of the three goals. For instance, when regulators focus on market protection, financial stability, and simple and clear rulemaking, they most likely will impose broad prohibitions

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that could greatly deter Fintech innovations. On the other hand, as regulators focus on promoting Fintech innovations and offer simple rules, they would have to apply an easy and low-intensity regulatory agenda that could not guarantee protection, reliability, and solidity in the financial system. Also, as regulators focus on empowering innovations and stimulating market integrity, they would probably enforce more complicated laws that are less clear and not simple to understand. In addition, as the prospects of Fintech evolve with the strong presence of Big Data and machine learning techniques, the present trend of Fintech has created informational and regulatory gaps that require attention to be addressed.

Moreover, technological developments have heightened the uncertainties and complexities as well in estimating the influence of new innovations on the customer, market, and financial industry in general. Hence, many governments have started to review their regulatory frameworks so that it does not restrain innovation and market entry [34]. This is especially significant in developing market economies, where efficiency and entry advantages are exceptionally substantial. Thus, many countries have adopted innovation enablers, such as regulatory sandboxes that test new innovative products in a controlled setting. Jagtiani and John have provided an analysis of Fintech technologies and their regulatory concerns, especially focusing on consumer protection on account of Fintech development. Also, the literature offers a set of strategies to regulate Fintech's possible threats. These strategies include informal supervision, no-action letters, regulatory sandboxes, experimental programs, and licensing as opposed to chartering organizational structure [33].

Fintech regulatory sandboxes are regulatory pioneering approach in Fintech that depends on informal guidance [33]. In particular, the sandbox is a method through which Fintech firms can conduct tests on new Fintech products and services within a living setting, and, more importantly, under a regulatory supervision, where they are subject to specific restrictions and safeguards. Within the sandbox, innovators would be offered an environment where they could conduct experiments and test their new designs [33]. Mainly, the value proposition of Fintech sandboxes basically entails an exchange, where the sandbox applicant can benefit from lower barriers to entry into a regulated market for narrow access to that market in return. Furthermore, well-governed sandboxes can support regulators' double mandate of protecting customers by inhibiting fraud and decreasing market distortions [35]. The relaxed regulatory setting of the sandboxes gives Fintech firms more room for testing their innovations without causing harm to the general consumers or the financial system [33]. For instance, a regulatory sandbox for robo-advising was proposed in the literature in which participants can experiment with their robo-advising products inside the actual market, interacting with real customers, although these products will be under close observation of the administrator [33].

In general, regulatory sandboxes present many benefits, including creating an approachable general regulatory system for Fintech services, decreasing the regulatory ambiguity associated with Fintech, and improving its regulatory and administrative capacity [33]. For instance, some studies have highlighted the impact of the absence of transparency in which unconventional data attributes may be applied in credit risk assessment resulting in regulatory uncertainty. A sandbox can allow the formation of shared knowledge for both the Fintech firms and the regulators that could reduce regulatory ambiguity for all the market participants. Hence, Fintech regulatory ambiguities and the absence of regulations transparency have become a critical concern in the Fintech industry as they establish their policies and business models [33].

III. Methodology

A. Hierarchical Decision Model Overview

This research aims to highlight the key factors that impact the effectiveness of policy instruments on increasing the adoption of Fintech by SMEs by developing a comprehensive decision model. The methodology used in this research is the HDM. HDM, a multicriteria method, has been used in a wide range of applications, such as selection, assessment, planning and development, significance and ranking, decision-making, and forecasting. More importantly, the HDM methodology has been previously applied by research and proved its viability in assessing policy instruments effectiveness in different industries [36], [37]. The HDM was first introduced by Cleland and Kocaoglu [38], and ever since it has been broadly used as a decision-making tool for different purposes, especially for analyzing and assessing technology adoption in various fields such as healthcare, transportation, energy, and innovation [39], [40]. HDM process consists of several steps including creating the decision problem and dividing it into levels, followed by pairwise comparison among all variables, calculating the priorities of the objectives, and checking the consistency [41]. Hence, an HDM model will be developed and used in this research to achieve its objective, where it will offer a policy assessment tool for policy makers to evaluate the effectiveness of available policy instruments and create the ideal portfolio of Fintech policy on the basis of multiple circumstances and scenarios.

B. Desirability Curves Metrics

Using desirability curves along with the HDM model helps in identifying the desirable levels/metrics for each decision factor. The approach of using desirability curves as a supplement for the HDM model was first introduced by Phan [42]. As the experts quantify the model's parameters, they do the same for the desirability metrics. Experts' judgments will remain constant, while different policy instruments will be examined against the experts' results using the instrument's performance level in terms of each metric to assess their effectiveness in increasing the adoption of Fintech on the desirability metrics scale. In this research, desirability curves will be applied to identify the desirability or importance of each metric for the decision maker. Hence, experts will be presented with the units of measurement and their categories for each factor in the model. Each factor in the model will have a set of metrics representing the possible performance levels of that factor with regard to the model's objective. The experts will assign a value ranging from 0 to 100 to each metric establishing the foundations of how desirable the

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metric is. Then, the curves will be illustrated based on the calculated averages of the experts' assessments.Figs. 2–18 show the desirability curve for one of the model's factors. Furthermore, the effectiveness of the policy instrument on the adoption will be calculated by multiplying each factor's weight by its desirability value, as demonstrated in the following section [36], [39], [43], [44], [45].

C. Calculating the Instrument's Effectiveness Score

Through the expert's subjective judgments, the effectiveness score of Fintech policy instruments will be measured. Experts will help in determining the relative significance of the policy targets and perspectives to the research objective. Furthermore, the effectiveness of the policy instrument on the adoption will be calculated by multiplying each factor's weight by its desirability value, as demonstrated in the following equation [36], [39], [44], [45]:

$$\text{Instrument Effectiveness Score} = \sum_{p=1}^{P}\sum_{c=1}^{C} P_p\, C_c^p\, d\left(m_{p,\,cp}\right)$$

where

Instrument Effectiveness Score = Fintech policy instrument effectiveness score for increasing Fintech adoption by SMEs;

C	number of criteria;
P	number of perspectives;
P_p	degree (weight) to which perspective p contributes to the mission;
C_c^p	relative contribution of the criterion c below perspective p in relation to the mission;
$d(m_{p,cp})$	desirability value of the performance metric for the cth criterion under the pth perspective.

D. Experts Judgment

The use of experts' judgments has been considered as one of the important advantages of the HDM approach. The input of subject matter experts is essential for the validation and quantification of the research model. However, the model results' validity and reliability are very dependable on a good formation of the expert panel and inviting the right experts to participate in the research process. During this research, experts are requested to offer their opinions from different fields and for various reasons, for instance, validating a research's results or identifying serious concerns associated with a particular topic. The duration that takes an individual to achieve the point where he/she could be considered an expert is immense. Ericsson et al. [46] claimed that "it takes time to become an expert. Even the most gifted performers need a minimum of ten years of intense training before they win international competitions." An expert's judgment on a particular subject has been defined as "data given by an expert in response to a technical problem" [46]. Subject matter experts experienced in the field of Fintech and Fintech-related policies will validate the model's perspectives, criteria under each perspective, desirability metrics, and the outcomes of the model's quantifications. It is important for the sake of the model's robustness and generalizability that the experts involved in validating the model have a thorough knowledge and solid background in Fintech and SMEs, diverse backgrounds, and different sorts of experience. Experts for this research are identified using several approaches: social network analysis, personal connections from the field of Fintech, snowballing sampling, and reports of technology and governmental organizations and academic papers. This research utilizes inputs from 55 subject matter experts and those experts are distributed across 12 expert panels to perform the validation and quantification of the model constructs. The expert's panels validate the main model perspectives and each perspective with its associated factors. Experts will quantify the main model perspectives and each perspective associated with its factors as well. Also, the dedicated panel will be assigned to validate and quantify the desirability curves and evaluate the overall results of this research. The experts are involved throughout this research.

E. Inconsistency and Disagreement

The HDM results will be analyzed for inconsistency and disagreement. Inconsistency arises if there is a conflict within a single expert's evaluations, where the expert's pairwise comparison judgments are inconsistent [45]. On the other hand, disagreements arise between experts when different quantifications and different perspectives on the same analysis are given by the experts within a single panel [47], [48], [49]. Since experts are expected to be from diverse backgrounds and hold different values, experiences, and interests, disagreements are common to exist between experts' judgments. The acceptable expert judgment inconsistency and disagreement between experts is 0.10 threshold, as previously applied by various studies [50], [51].

F. Methodology Justification

The literature highlights the valuable role of policy instrument assessment in supporting governments to recognize and enhance the way these instruments work in order to increase their effectiveness, validate the outcomes and effects of these instruments, and select the instruments that are most suitable for their country [52]. Furthermore, the literature emphasizes the importance of a well-established multicriteria evaluation method as a valuable tool for assessing policy makers in conducting detailed and systematic policy evaluations using criteria that demonstrate their preferences and allow stakeholders to be involved. More importantly, evaluating policy instruments can instruct policy makers on how the implemented policy can be enhanced, as it can advise governments on the measures that need to be taken to improve or modify available policies in order to improve their effectiveness. Hence, policy assessment ought to be a critical information source for future policy making and for preparing the policy agenda as well.

In general, there are various methods that support policy evaluation research ranging from qualitative to quantitative analysis. Typically, the qualitative research approach concentrates on the process and highlights the performance of the different actors engaged in the assessment. It tends to be more flexible and could lay the results in their social and political framework. On the

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TABLE I
RESEARCH PERSPECTIVES

Perspective	Objective	Details
Organizational Perspective	Improve managerial skills and efficiency in SMEs	When making a policy that targets SMEs, it is important to take into consideration organizational factors that lay within the SME and can relate directly to the adoption decision. This perspective measures the importance of the SME's management in facilitating the adoption of Fintech [58] [59] [60].
Technical Perspective	Assess SMEs digital capabilities	Understanding the needs and challenges that SMEs face with technical development is important for improving current policies and for future policy planning. This perspective measures the importance of developing the technical capabilities for Fintech adoption [61] [62] [63].
Economical Perspective	Broaden financing options for SMEs and provide better financial support	For SMEs, finance is a vital aspect. The availability of the right financing options along with efficient financial management is very important for the survival of firms, especially SMEs. This perspective measures the importance of economic factors for SMEs adoption of Fintech [63] [64].
Regulatory Perspective	Enhance regulations and compliance on Fintech to lower associated risks	This perspective deals with regulatory issues associated with Fintech adoption. The objective is to form a policy that is easy to employ, add value, and ensures consumer protection. This perspective measures the importance of considering general performance of Fintech policy to increase Fintech adoption by SMEs [34] [65] [66] [67].

other hand, qualitative research methods could provide a reasonable justification of the reasons a specific policy involvement has or has not had an impact on a particular situation. However, quantitative and qualitative research methods are complementary and not interchangeable, where they can balance each other in a given research and strengthen its outcomes. For instance, a quantitative tool can offer a statistical foundation for a more qualitative method, whereas a qualitative tool can demonstrate the connections created through quantitative research. Both a theoretical basis and a qualitative assessment of the processes causing presumed relations are essential for a statistical analysis substance. Hence, because the HDM model, an MCDM approach, offers both qualitative and quantitative analysis, it would be a suitable and comprehensive approach for conducting this research on policy instruments evaluation.

The MCDM research approach is a branch of operation research models and a renowned discipline of decision making [53]. In general, MCDM methods could be classified into two groups: multiobjective decision making (MODM) and multiattribute decision making (MADM) [54]. In MODM, the decision-making subject is analyzed by multiple and competitive objectives that need to be improved based on a range of possible and available impediments, while in MADM, a range of alternatives is assessed by being compared to a set of criteria. The MADM approach is one of the highly used MCDM methods applied to resolve complicated problems. MCDM encompasses many distinct methods, especially the analytic hierarchy process, preference ranking organization method for enrichment evaluation, elimination and choice translating reality method, HDM, and multiattribute utility theory [36].

It has been argued that because the human brain is incapable of handling too many factors simultaneously, it tends to analyze complexities by sorting them and dividing their components into smaller elements instead of dealing with each element separately. Hence, this vertically hierarchical composition is the ecological way of human thinking. Also, it has been proven that a cross-sectional approach of evaluating associations is advantageous, especially when there is a specific objective and the impact of additional factors or the connection between entities needs to be understood [36], [55]. Thus, the HDM model enables decision makers to split the problem into smaller entities to be explored and identify any hidden associations between the problem's elements. More importantly, the HDM methodology has been used for policy planning and evaluation for a variety of objectives and was proven practical [36], [56], [57].

Furthermore, traditional decision-making methods can no longer deal with the density of current systems when dealing with such problems. Hence, an MCDM approach offers an adaptable method that can synthesize and evaluate a broad variety of attributes in multiple ways and give valuable insights to the decision maker by breaking down the problem. Also, MCDM methods can deliver a practical scientific decision-making support instrument that can rationalize its preferences in a clear and reliable manner. More importantly, most policy evaluation approaches evaluate the adoption from a limited perspective merely a technical one, while the MCDM method can analyze the problem with an inclusive lens by identifying and including the most important perspectives that are involved in the decision and policy making. This thorough approach can enhance the effectiveness and reflection of the policy with regard to stakeholders' concerns in order to increase their adoption. Also, MCDM methods would offer a robust tool for ranking multiple alternatives relative to policy instruments for Fintech adoption in the existence of various objectives and constraints. Hence, it can offer informative recommendations relative to existing policies for possible modifications to increase their effectiveness. Also, through incorporating desirability curves, policy makers would have a tool that sets a score for any given policy instrument evaluated, and based on this score, they can decide what aspects to alter or emphasize in order to increase the instrument's overall effectiveness over the adoption.

IV. MODEL DEVELOPMENT

A. Initial HDM Model

1) Model Perspectives: Through the literature review, four main perspectives were identified as important factors for a comprehensive evaluation of Fintech policy instruments. Table I illustrates each perspective with its objective and description.

2) Model Criteria: Tables II–V list the identified criteria (policy targets) with respect to the research goal.

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Organizational Perspective Criteria

TABLE II
ORGANIZATIONAL CRITERIA DEFINITIONS

Criteria	Description	References
Management Awareness and Trust	SMEs management's lack of knowledge and awareness about the available financing options and alternatives is a fundamental barrier. This variable measures the importance of management awareness and trust in SMEs adoption of Fintech.	[68] [69] [70]
Innovativeness Encouragement	Innovativeness is the willingness to discover new technologies and is eager to experiment with Fintech services. This variable measures the importance of encouraging innovativeness in the adoption of Fintech by SMEs.	[64] [65] [71] [72]
Open Innovation Participation	Public policy can efficiently facilitate and effectively increase collaborative innovation for SMEs. This variable measures the importance of open innovation participation in the adoption of Fintech by SMEs.	[73] [74]
Discrimination and Bias Elimination	SMEs face internal and market challenges that include insufficient financial skills and lack of knowledge and awareness about financing options. However, these challenges are typically more noticeable in certain segments of the business population, particularly firms led by groups that are under-represented in entrepreneurship. This variable measures the importance of eliminating discrimination and bias on SMEs adoption of Fintech.	[68]
SMEs Benefits Prioritization	Studies show that SMEs' adoption of Fintech is directly impacted by their perceived usefulness of Fintech services. This variable measures the importance of prioritizing SMEs benefits on the adoption of Fintech by SMEs.	[49] [58]

Technical perspective criteria.

Technical Perspective Criteria

TABLE III
TECHNICAL CRITERIA DEFINITIONS

Criteria	Description	References
Security	SMEs using Fintech are faced with threats of fraud, information leaks, and cyberattacks making security. Addressing such issues is a policy priority. This variable measures the importance of enhancing security in SMEs adoption of Fintech.	[70] [72] [75] [76]
Digital Infrastructure Improvement	A digital infrastructure should be affordable, promote widespread internet access and use, and foster the use of new technologies. This variable measures the importance of improving the digital infrastructure in the SMEs adoption of Fintech.	[70] [77]
Privacy and Data Protection	Adequate data protection and data privacy regulations are key for protecting consumers' interests and building trust. This variable measures the importance of supporting privacy and data protection in the adoption of Fintech by SMEs.	[58] [77] [78]
Digital Literacy	A universal constraint for SMEs is their insufficient understanding of digital technologies and their lack of skills necessary for digital adoption. This variable measures the importance of digital literacy in SMEs adoption of Fintech.	[77] [68]
Information Asymmetry Prevention	Information asymmetry happens when information related to the borrower's credit, project risk and benefits are known by borrowers more than by financial institutions. This variable measures the importance of preventing information asymmetry in SMEs adoption of Fintech.	[77] [79] [80]

Economical perspective criteria.

Economical Perspective Criteria

TABLE IV
ECONOMICAL CRITERIA DEFINITIONS

Criteria	Description	References
Diversify SMEs Financial Channels	Policy makers can support diversifying financing models and growing risk appetites in the market for SME financing. This variable measures the importance of diversifying SMEs' financial channels on the adoption of Fintech by SMEs.	[68] [79] [80]
Cost Saving	Digitization and automation enable the financing process to become more efficient, thereby, lowering costs. Cost reduction influences SMEs' growth. This variable measures the importance of cost saving on the adoption of Fintech by SMEs.	[60] [81] [82] [83]
Financial Literacy	The lack of sufficient financial skills and knowledge makes it difficult for SMEs to adopt Fintech. This variable measures the importance of enhancing financial literacy on SMEs adoption of Fintech.	[68] [84] [85]
Financial Efficiency	Financial efficiency occurs when the necessary requirements (i.e., elimination of market distortions, competitive markets, accessible information, etc.) are met to provide the highest quality financial services at the lowest cost possible. This variable measures the importance of financial efficiency on the adoption of Fintech by SMEs.	[60] [81] [83]

Regulatory perspective criteria.

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Regulatory Perspective Criteria

TABLE V
REGULATORY CRITERIA DEFINITIONS

Criteria	Description	References
Regulations Existence and Enforcement	Financial regulators must develop new regulation approaches to ensure financial stability and consumer protection. Regulations have been reported as a barrier to SMEs' adoption of Fintech products. This variable measures the importance of the exitance and enforcement of regulations on the adoption of Fintech by SMEs.	[71] [81] [86] [87]
Law Compliance	With many regulators involved, it can be difficult for Fintech firms to identify what laws are applicable to them and how their activities will be regulated. This variable measures the importance of Fintech firms' compliance with laws on the adoption of Fintech by SMEs.	[72] [86]
Policy Ease of Application	The ease of policy application and ability to work in the current policy system are important for policy effectiveness. This variable measures the importance of the policy's ease of application on the adoption of Fintech by SMEs.	[36]
Consumer Protection	Regulators aim to assure consumers protection from the procedures of businesses under their purview without taking actions that might impair market competition. This variable measures the importance of consumer protection on the adoption of Fintech by SMEs.	[71] [88]

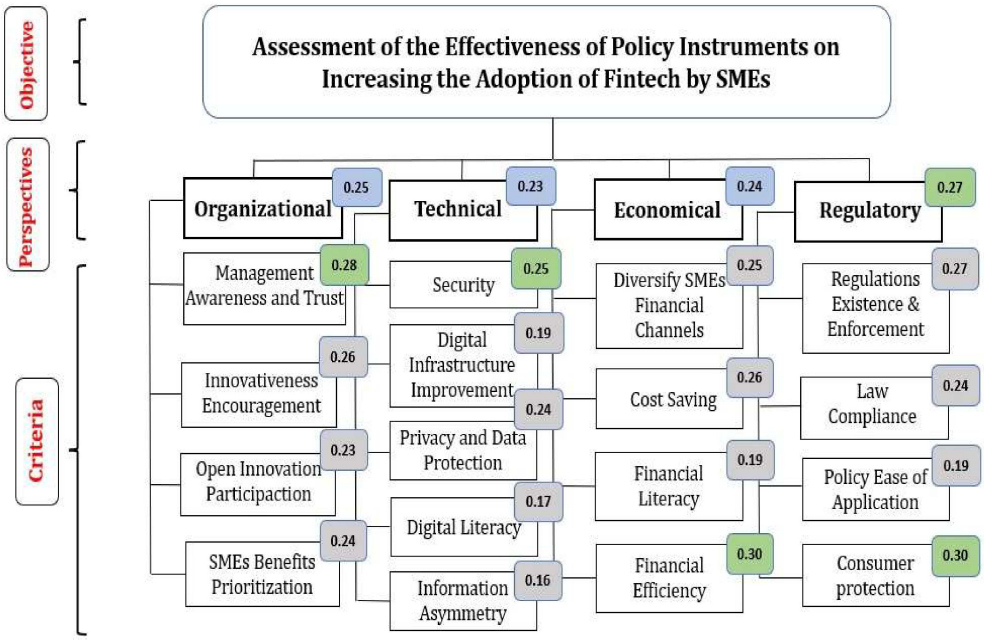


Fig. 2. Final HDM model with weights.

B. Data Collection

Experts were invited to participate in the validation and quantification of the research model. This research utilizes inputs from 55 subject matter experts distributed across 12 expert panels for validation and quantification.

1) Model Validation: Five expert panels were assigned to validate the research model parameters through a validation survey. In the validation phase, if the majority (two/third) of the experts approve all the components of the model, then the model is considered validated. Nevertheless, elements that fail to reach this threshold have to be eliminated from the model. In addition, if a new element was suggested by more than three experts, it has to be added to the model and the model will be validated once more. A total of 17 experts have been part of the validation phase. Based on the expert's validation, feedback, and discussion, some changes were applied to the initial model. Specifically, the Discrimination and Bias Elimination criterion was only approved by half of the experts in the panel. Since the approval was less than the threshold of two-thirds of the panel, the criterion was removed from the initial model. The finalized validated and quantified model is shown in Fig. 2.

2) Model Quantification: In this phase, the expert's subjective judgments are expressed in pairwise comparisons with a constant-sum measurement scale (1–99 scale) for comparing each pair of decision factors by the experts [47], [89]. The assigned experts quantified the model's elements to identify their relative importance to the overall objective of evaluating the policy instruments' effectiveness in increasing the adoption of Fintech by SMEs. The results of the perspectives quantification indicate that the regulatory perspective is the most important

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TABLE VI
MANAGEMENT AWARENESS AND TRUST QUANTIFIED DESIRABILITY CURVE

Management Awareness and Trust	
Description	**Desirability**
No level of impact on enhancing SMEs management awareness and trust	2
Low level of impact on enhancing SMEs management awareness and trust	25
Medium level of impact on enhancing SMEs management awareness and trust	50
High level of impact on enhancing SMEs management awareness and trust	74

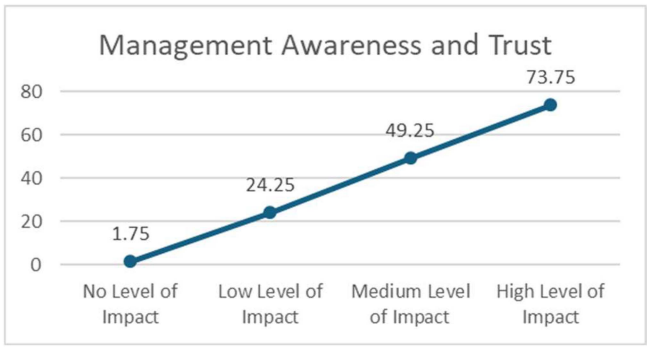


Fig. 3. Management awareness and trust quantified desirability curve.

perspective in evaluating the policy instrument's effectiveness in increasing Fintech adoption by SMEs, followed by the organizational, economical, and technical perspectives. The inputs from the experts were tested for inconsistencies and disagreement, which in this study did not exceed the threshold of 0.10 acceptable level. Fig. 2 displays the finalized model with the corresponding factor's weights.

3) Desirability Metrics: The objective of using desirability curves is to examine how desirable a metric is for the decision maker. Each desirability metric/level is assigned to a policy instrument performance against each factor in the model. The judgments of experts have been utilized throughout the validation and quantification process to develop the desirability levels. The experts assigned a score from 0 to 100 to each level within every single factor expressing how desirable that specific level is in terms of the policy instrument's performance. The desirability curves combined the model weights to calculate the effectiveness of the policy instruments score. Figs. 4–20 show the desirability curves for all the model factors. For example, the management awareness and trust factor measure the level of impact of the policy instrument on enhancing the awareness of SMEs management and bridging their trust gap in Fintech and digital finance ecosystem caused by the ambiguity associated with unfamiliar technologies and concerning data breaches that led to consumer's hesitation to adopt Fintech. The desirability curves for this factor can be presented in Table VI.

V. CASE STUDY

To ensure the robustness and reliability of the research model, this research will be applied to a real-world case study to test and assess the effectiveness of policy instruments on increasing Fintech adoption at the Saudi Arabian regulatory sandbox.

A. Fintech Regulatory Sandbox

To respond to the rapid advancements in the Fintech industry, regulators around the world have been developing regulatory sandbox regimes [90]. Many countries have adopted innovation enablers, such as regulatory sandboxes that test new innovative products in a controlled setting [33]. The main purpose of creating regulatory sandboxes is to help regulators and policy makers to achieve critical policy targets, such as consumer and investor protection, market integrity, financial inclusion, risk mitigation, and more importantly, promoting innovation [91], [92]. In addition, a regulatory sandbox allows Fintech companies to experiment with their products, services, solutions, and delivery mechanisms in the market inside a more relaxed governing environment, yet within a clearly defined space and duration agreed on with regulators, without directly facing all the typical regulatory consequences of engaging in such activities [93], [90]. Sandboxes permit Fintech companies to start their solutions at a lower cost, more effectively, and with higher quality. Moreover, a sandbox permits Fintech firms to make their platform accessible to a restricted number of consumers [93]. As a safety net, financial advisers would evaluate the platform once it is issued and before the transactions get executed.

The Saudi Arabian Regulatory Sandbox

In recent years, the Kingdom of Saudi Arabia has been undergoing substantial economic development transformations positioned around the country's national economic development strategy, the Saudi Vision 2030 [94]. One of the priorities of the 2030 Vision is advancing the financial services sector and digital transformation developing the country's digital economy, fostering and supporting innovation and entrepreneurship culture, and increasing the contribution of SMEs to the economy [95]. Saudi Arabia has developed a regulatory sandbox to support the global Fintech regulatory initiatives [96]. The goal of the regulatory sandbox is to understand and assess the impact of new technologies in Saudi's financial services market, as well as to help transform the Saudi market into a smart financial center.

For the case study, we assess the effectiveness of policy instruments on increasing Fintech adoption at the Saudi Arabian regulatory sandbox to present the effectiveness score based on the performance of the Saudi Arabian regulatory sandbox on the developed model, propose various enhancements, and suggest

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TABLE VII
SAUDI ARABIAN REGULATORY SANDBOX ASSESSMENT SCORE AND SUGGESTED IMPROVEMENTS

Perspective	Factor	Global Weight*	Value Curve Score	Original Score	New Curve Score	New Score	Actions / Recommendation
Organizational Perspective	Management Awareness and Trust	0.069	60	4.160	60	4.160	No change
	Innovativeness Encouragement	0.064	70	4.474	70	4.474	No change
	Open Innovation Participation	0.059	75	4.386	75	4.386	No change
	SMEs Benefits Prioritization	0.059	75	4.433	75	4.433	No change
Technical Perspective	Security	0.057	50	2.866	80	4.560	Improve security standards and enforce more strict guidelines
	Digital Infrastructure Improvement	0.043	90	3.832	90	3.832	No change
	Privacy and Data Protection	0.055	40	2.207	80	4.400	increase privacy measures and ensure consumer data is highly protected
	Digital Literacy	0.039	60	2.331	60	2.331	No change
	Information Asymmetry Prevention	0.037	50	1.863	70	2.590	Decrease information asymmetry by setting new rules that allow consumers to benefit from alternative data.
Economical Perspective	Diversify SMEs Financial Channels	0.060	90	5.394	90	5.394	No change
	Cost Saving	0.062	10	0.619	60	3.720	Encourage Fintech firms to be cost effective in order to allow SMEs to borrow money at lower rates or invest their money with higher returns.
	Financial Literacy	0.047	75	3.534	75	3.534	No change
	Financial Efficiency	0.073	70	5.129	70	5.129	No change
Regulatory Perspective	Regulations Existence and Enforcement	0.075	90	6.667	90	6.667	No change
	Law Compliance	0.065	50	3.244	70	4.550	Focus higher attention on educating Fintech firms on the current laws and regulations and enforce their compliance by prosecuting fraudulent actions.
	Policy Ease of Application	0.052	50	2.577	65	3.380	Reduce the ambiguity and complexity of the policy instrument so it could be applied easily and seamlessly.
	Consumer Protection	0.084	50	4.174	90	7.560	Add more regulations that guarantee consumer protection and safeguard their interests.
			Score	**61.89**	**Improved Score**	**75.10**	

Global Weight*: the global weights present the relative weight of the factors with respect to the overall objective. It shows the importance of the factors against each other in the model.

recommendations. Table VII lists the final scores and proposed recommendations for improvements.

Besides presenting the current state of the effectiveness of their policy instruments, this research proposes a set of recommendations for improvements. This research assists policy makers in evaluating the effectiveness of their policy instruments in increasing the adoption of Fintech by SMEs, identifies and ranks the most important factors to the adoption, as well as detects the weaknesses that could negatively impact the policy instrument's effectiveness. Hence, corrections and restorative measures can be taken depending on the identified weaknesses. Table VII lists the existing performance and suggested enhancements and their impact on the effectiveness score. The suggested improvements are directed toward the areas that have scored low and provide suitable suggestions accordingly. Policy makers can undertake the changes either conservatively, moderately, or extremely. The policy maker may turn to value curves to locate the policy instrument's current state and identify what the following level for each model's element is and decide what would be the ideal level for each element and apply the suggested recommendations to reach it.

VI. DISCUSSION

A. Practical Application: General

While literature has emphasized the significant role of government support in accelerating Fintech adoption by SMEs [96], Yet, most of the available literature on Fintech concentrates on Fintech service management and innovation. Also, most of the available Fintech adoption models are limited in their scope. In fact, currently, there are no models or tools designed to evaluate and assess the impact of the available policy instruments in increasing SMEs adoption of Fintech that combine multiple perspectives to evaluate and measure the instruments' effectiveness.

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Accordingly, 17 factors have been validated by subject matter experts to be among the most critical considerations that might affect SMEs' decision of adopting Fintech and a policy maker should take them into consideration when designing a policy instrument that aims to increase SMEs' adoption of Fintech.

In addition, the experts' quantifications of the model have generated notable outcomes. First, the results show that the technology perspective has ranked the lowest among the four main perspectives with a relative weight of 23%. This perspective contains the technological considerations and concerns related to Fintech. According to the experts, technological factors are not the main impediments to SMEs' adoption of Fintech. Although Fintech encompasses intercrossed domains, including technology, which increases its complexity, SMEs have concerns that are greater than technological factors when it comes to their adoption decision. Thus, the HDM model presented in this research adds value by helping policy makers to comprehensively evaluate the effectiveness of their policy instruments against each critical factor and learn how to improve the effectiveness in order to generate higher adoption rates.

On the other hand, an important insight from the model's quantifications results was that the regulatory perspective ranked the highest among the four main perspectives with a relative weight of 27% based on the experts' judgments. This perspective encompasses the governing aspects that concern an SME when it comes to Fintech and, therefore, may impact their adoption decision. This ranking indicates that the factors under the regulatory perspective play an important role in SMEs adoption of Fintech. More importantly, it shows that the more the policy instrument satisfies the policy targets under this perspective, the higher its effectiveness becomes in increasing Fintech adoption. Denoting that SMEs need to trust that the governing authorities of Fintech will protect their interest and eliminate the risks associated with them, and their adoption of such technologies will only add benefit to them.

B. Practical Application: Organizational

From the organizational perspective, four factors have been validated by the experts as important. These factors are: Management Awareness and Trust, Innovativeness Encouragement, Open Innovation Participation, and SMEs Benefits Prioritization. The quantification results show that Management Awareness and Trust is the most important factor from the organizational perspective with a relative weight of 28%. In fact, it has been argued in the literature that the failure of many small businesses was not tied to the owner's incapability or the provision of inferior services; however, it was mostly because they did not run their firms as business [97]. Most SMEs owners lack sufficient knowledge for success, especially knowing how to manage and expand their businesses. More importantly, research that has studied technology adoption in SMEs has highlighted the significant impact of the characteristics, behaviors, and attitudes of SMEs' owners/managers on explaining the variances in the level of innovation adoption [98], [99], [100], [101]. Therefore, increasing SMEs' awareness about Fintech and gaining their trust to adopt such technologies is the most important organizational factor that policy makers need to focus on. Furthermore, it has been proven that promotion is very important in establishing awareness about online services [102]. Therefore, the policy instrument framework should aim to promote Fintech activities in continuous and efficient ways. It should also promote the benefits Fintech offers to SMEs (i.e., reduced transaction costs, improved cash flow, increased competitiveness, etc.) [103]. Accordingly, special promotional tools for SMEs are exceptionally needed since they require a unique approach compared to those of individual consumers.

C. Practical Application: Technical

In addition, the experts have validated five factors from the technical perspective as important. These factors are: Security, Digital Infrastructure Improvement, Privacy and Data Protection, Digital Literacy, and Information Asymmetry. Through experts' quantifications, Security was identified as the most important technical factor with a relative weight of 25%. This outcome did not come as a surprise since it has been argued in the literature that security may possibly be a major barrier to technology adoption and that information safety is vital for the integrity of the technology [104], [105], [106]. Hence, better action should be directed toward security, especially since it is linked to the safety of the users of the entire emerging Fintech technology. In Fintech literature, security has been found to be among the most common factors that influence customer satisfaction, and the assurances of users' safety and confidentiality of the data generated during their online transactions process were of greatest importance [107]. Policy makers could set high standards for security quality assurance in their regulation of Fintech solutions. Studies show that improper handling of personal data and low transparency and ethics can cause massive risks, and one way to handle that threat is by issuing high-level standards for Fintech companies to follow and, hence, control those risks [108].

D. Practical Application: Economical

Moreover, experts have validated four factors from the economical perspective, including Diversify SMEs Financial Channels, Cost Saving, Financial Literacy, and Financial Efficiency. The experts have identified Financial Efficiency as the most important Economical factor with a relative weight of 30%. In fact, it has been argued in the literature that Fintech has the ability to increase SMEs odds of borrowing loans at lower interest rates [109], [110]. In addition, Fintech can accelerate the loan application process and assist borrowers to get funds faster, and, accordingly, improve SMEs competence in using the funds in a timely manner [111], [112]. Also, Fintech can aid SMEs in getting investment management consultation at a reduced cost [76], [77]. Consequently, Fintech could potentially impact SMEs revenues and costs, hence, their financial efficiency [77]. In addition, studies have shown that in order for SMEs managers to achieve greater efficiency, adopting Fintech could perform as a medium to realize this goal. However, culture has a significant impact on the relationship between Fintech and SME efficiency [68], [80]. Therefore, culture may need to be incorporated when formulating policies to support Fintech startups, as researchers have argued the importance of policy makers may provide advanced support to potential FinTech owners in individualistic or long-term-oriented cultures.

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E. Practical Application: Regulatory

Furthermore, the experts have validated four factors under the regulatory perspective as important. These factors include Regulations Existence and Enforcement, Law Compliance, Policy Ease of Application, and Consumer Protection. The Consumer Protection is ranked as the most important Regulatory factor with a relative weight of 30%. Advanced technology embedded in Fintech has offered great benefits to both consumers and providers [65]. However, the benefits of Fintech also indicate its potential to be disruptive and create new sorts of risk. Therefore, consumer protection has been an urgent priority for regulators around the world to preserve financial stability while creating a safe environment for Fintech innovations as well. Hence, the benefits of advanced technologies in Fintech make regulatory policy makers face a tradeoff between safeguarding consumers' interests and supporting the Fintech revolution and competition. Furthermore, as in any other financial regulation, there is a need for policy makers to coordinate inside the public sectors and among the public and private sectors to facilitate an interaction between the different Fintech regulations with the existing financial regulations or with other regulations that are relative to Fintech. Most importantly, it is vital for regulators to determine the most suitable means to develop and successfully apply cross-industry, cross-jurisdiction, and cross-country guidance approaches that offer consumer protection and preserve financial stability while fostering a safe environment for Fintech innovations.

VII. Conclusion

This research was built upon a comprehensive literature review in addition to the integration of the insights of knowledgeable and experienced subject matter experts, which resulted in an HDM model incorporating a set of perspectives and their criteria (policy targets) that impact SMEs adoption of Fintech. Through using the developed evaluation tool, the assessment of current policy instruments revealed that the regulatory perspective is the most important perspective with respect to increasing Fintech adoption. Accordingly, the best policy instrument should include policy targets that make Fintech regulations competent and inclusive to SMEs concerns in order to enhance Fintech solutions' reliability and, hence, accelerate the adoption. The relative importance/weights of the model's parameters were assigned by the judgments of experts. This research proposed a robust decision-making model, the HDM, to be used as a policy evaluation tool to assess the effectiveness of policy instruments on increasing Fintech adoption. To test its practicality, the developed model was applied to an existing policy instrument, the Saudi Arabian regulatory sandbox, where it has proven its capability in assessing the effectiveness. It also provides enlightening guidance and recommendations for possible improvements through the desirability curve levels and the discussion on the various tactics to achieve the improvements.

A. Research Contribution

The main purpose of conducting this research in the first place is to add and contribute to the assessment of disruptive innovations, Fintech, entrepreneurship and financial technologies adoption, policy evaluation, and financial inclusion from a policy standpoint through developing a holistic and robust decision making tool, the HDM, to help in policy evaluation and assessment in a quantitative and qualitative way. More importantly, financial technologies are considered a new and rapidly growing field; hence, it has not been thoroughly explored through sophisticated systematic academic research. Most available studies on Fintech and Fintech policies are limited in their scope. Also, the prospects and potential benefits of financial technologies have recently become an increasingly fascinating phenomenon, which explains why the literature available on this topic is fairly limited and new (i.e., most papers were published over the last two years). Thus, this research introduced a new theoretical framework that will be the first that offers a comprehensive examination of the Fintech adoption and includes the essential policy factors impacting Fintech adoption and assessing their weights in the context of SMEs. This research will advance the research on the perspectives and criteria key to the effectiveness of policy instruments targeting technology and innovation adoption. The model developed will advance decision and policy makers knowledge on the four main perspectives: organizational, technical, economical, and regulatory, and their deep effect on increasing the adoption of Fintech.

From a practical standpoint, this research will provide a comprehensive framework for policy makers to help simplify their understanding of the factors that affect the adoption of financial technologies by SMEs. It will present a useful evaluation tool that identifies the critical and challenging factors that need to be accurately accomplished by a policy instrument to enable and increase its effectiveness in increasing SMEs adoption of Fintech. This research will help policy and decision makers of Fintech and SMEs to classify and rank their priorities and improve their knowledge of where they currently are placed and where they should be instead in terms of policy target attainment. More effective policy instruments mean increased adoption of Fintech by SMEs, which could potentially result in increasing their financial inclusion by offering them better and more accessible financing options that will support their survival, and eventually contribute to the economic growth.

B. Limitations

There are possible limitations to this model, for instance, applying the model may or may not be the same for other regions or countries. Still, even when applied in other regions, the model will preserve the same structure that can be assumed to be valid in other settings with slight alterations. Also, the research model was mainly developed with consideration for SMEs adoption of Fintech. It may be applied to evaluate the adoption of Fintech by other entities; however, whether all the factors identified in this research would be applicable to other entities as they are to SMEs is something that cannot be determined conclusively by solely depending on this research. Moreover, the results of this research are context and time dependent. This implies that at any time in the future, the organizational, technical, economical, and regulatory factors may not be in the same condition as they were when this study was conducted. Precisely, weights and judgments of these factors could vary with time due to different reasons. In addition, this research applies the HDM methodology

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which relies heavily on inputs from subject matter experts. The results of the research depend on human insights, subjectivity of experts' opinions is inevitable. In addition, expert panels are assumed to be free of biases since experts' selection was dependent on their capability of offering valuable judgments and having sufficient knowledge of SMEs, Fintech, and its regulations.

C. Future Work

The previously discussed limitations open a starting gate for future research to clarify the remaining ambiguities and bridge the knowledge among relative topics. Thus, future research can investigate the effectiveness of policy instruments on individual forms of Fintech (i.e., mobile payments, money transfers, peer-to-peer loans, etc.) to compare how their outcomes may differ. In addition, since the case study analyzed in this research was the Regulator Sandbox policy instrument, new research can be developed using the proposed HDM model to evaluate additional policy instruments and examine their effectiveness on Fintech adoption by SMEs. Finally, an interesting prospect would be exploring the effectiveness of the Regulatory Sandbox in other regions to examine the performance of the policy instrument in that specific region. This could start an initiative for a guiding framework for policy makers to benchmark the instruments across countries and identify the similarities and differences to learn from past experiences.

APPENDIX
DESIRABILITY CURVES

A. Organizational Perspective Desirability Metrics

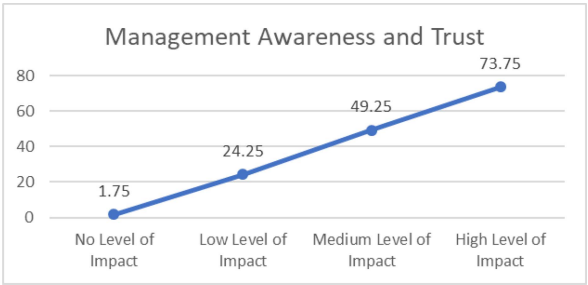


Fig. 4. Management awareness and trust quantified desirability curve.

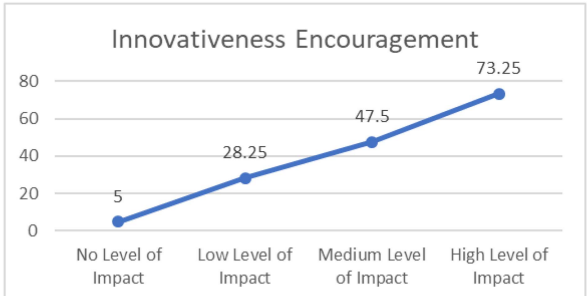


Fig. 5. Innovativeness encouragement quantified desirability curve.

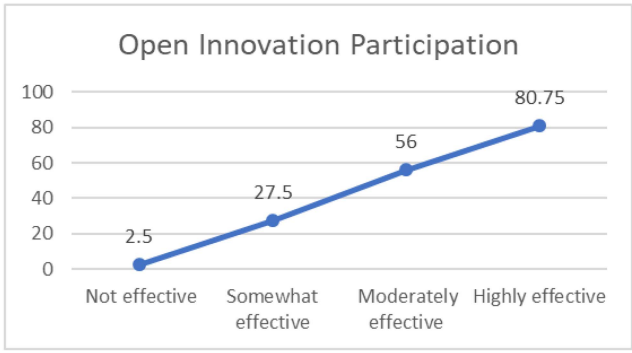


Fig. 6. Open innovation participation quantified desirability curve.

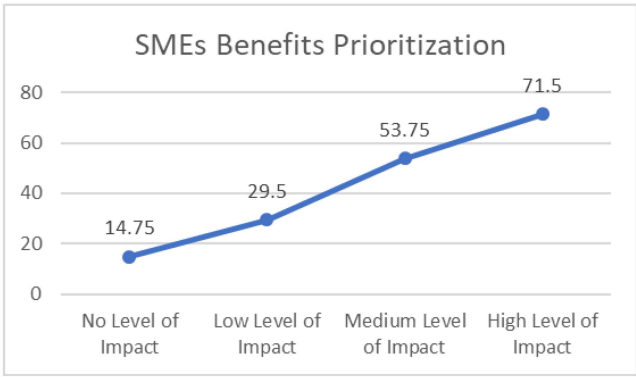


Fig. 7. SMEs benefits prioritization quantified desirability curve.

B. Technical Perspective Desirability Metrics

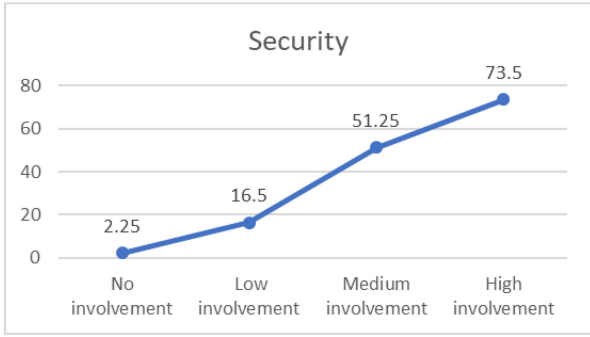


Fig. 8. Security quantified desirability curve.

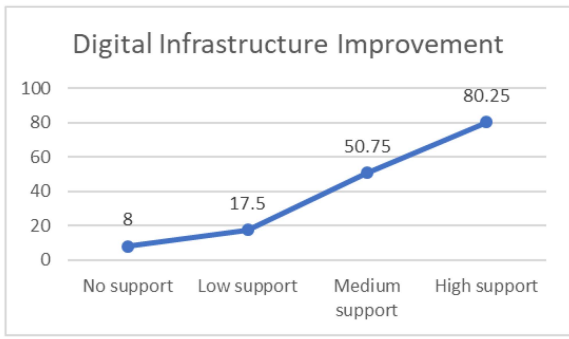


Fig. 9. Digital infrastructure improvement quantified desirability curve.

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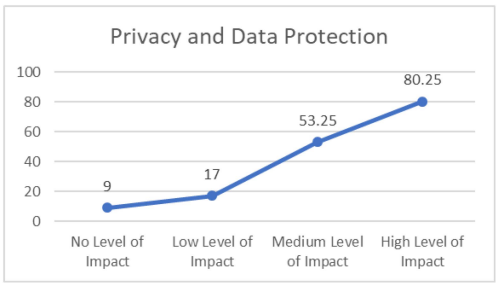


Fig. 10. Privacy and data protection quantified desirability curve.

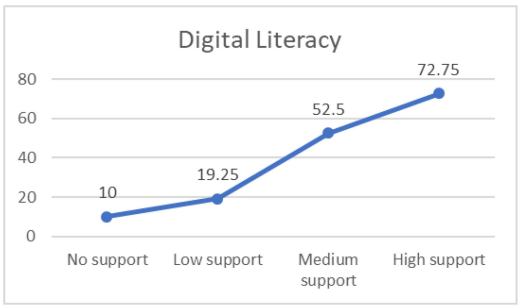


Fig. 11. Digital literacy quantified desirability curve.

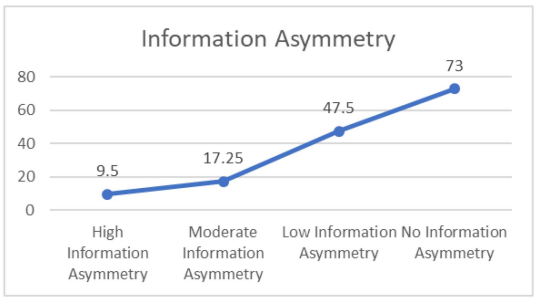


Fig. 12. Information asymmetry quantified desirability curve.

C. Economical Perspective Desirability Metrics

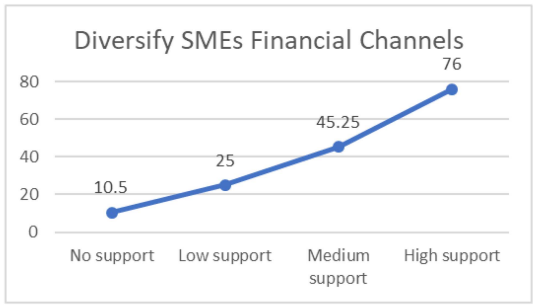


Fig. 13. Financial channels quantified desirability curve.

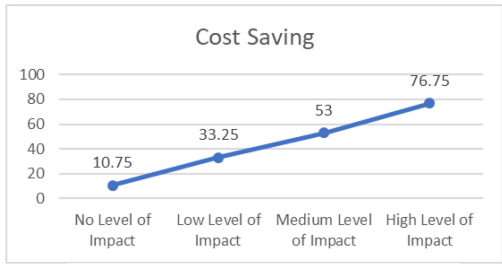


Fig. 14. Cost saving quantified desirability curve.

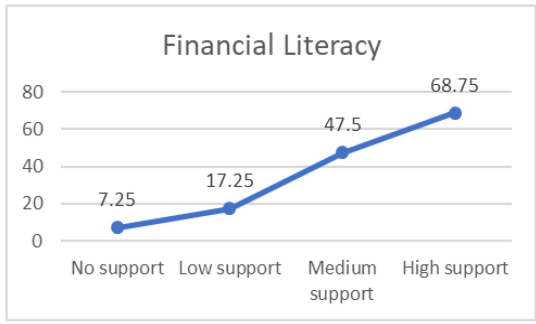


Fig. 15. Financial literacy quantified desirability curve.

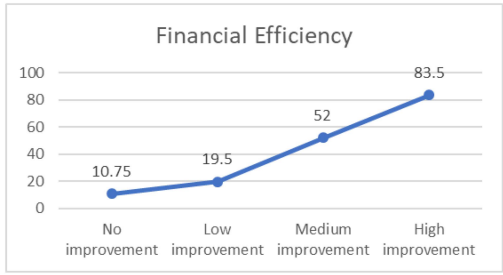


Fig. 16. Financial efficiency quantified desirability curve.

D. Regulatory Perspective Desirability Metrics

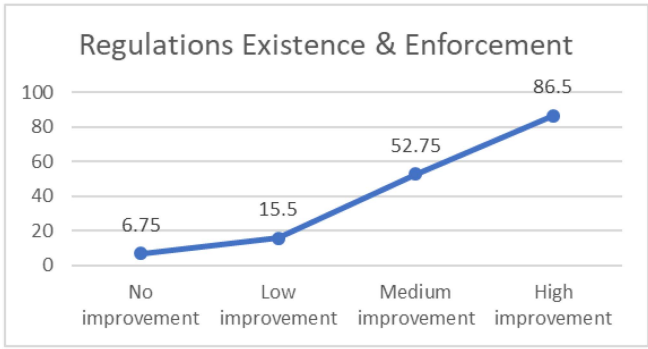


Fig. 17. Regulations existence and enforcement quantified desirability curve.

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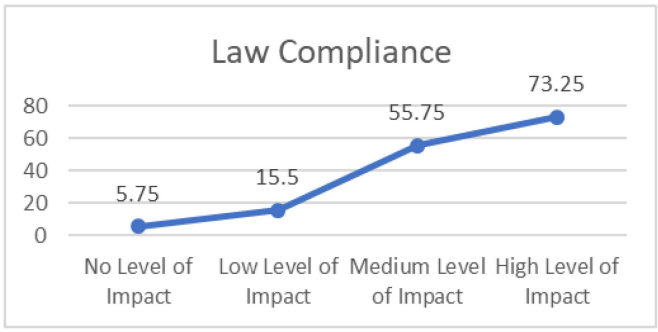


Fig. 18. Law compliance quantified desirability curve.

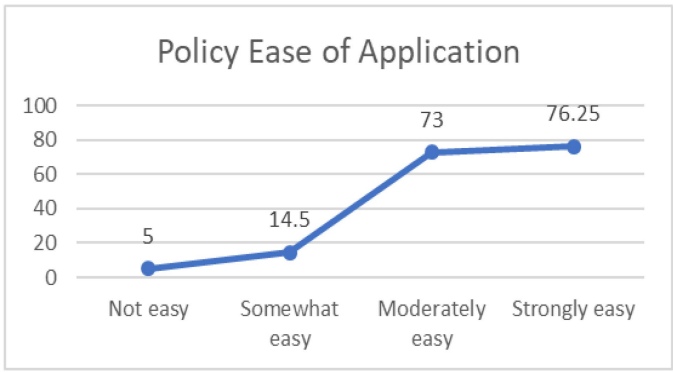


Fig. 19. Policy ease of application quantified desirability curve.

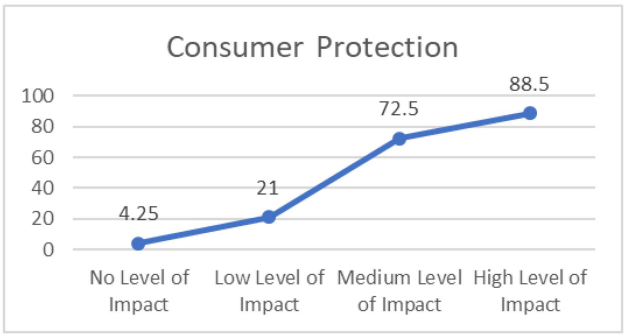


Fig. 20. Consumer protection quantified desirability curve.

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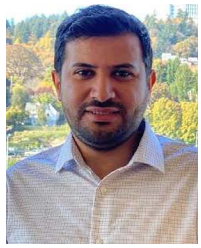
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